

Fender Vibroverb 6G16 Tremolo Circuit



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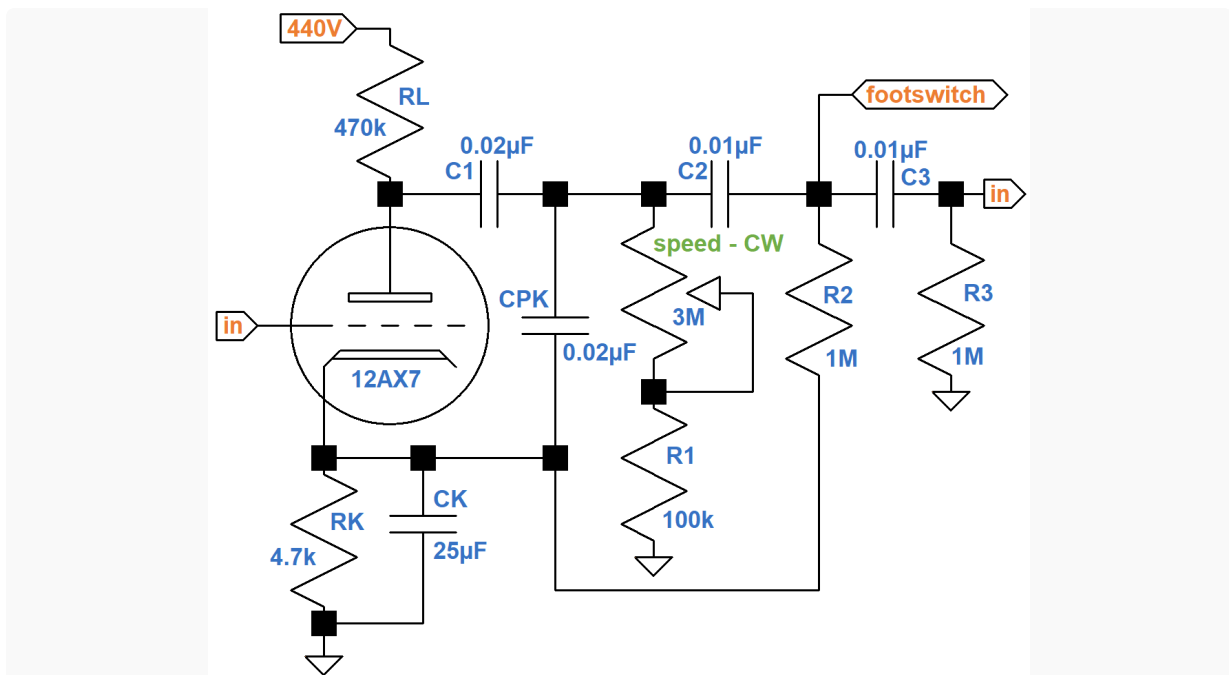
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The Low-Frequency Oscillator (LFO)

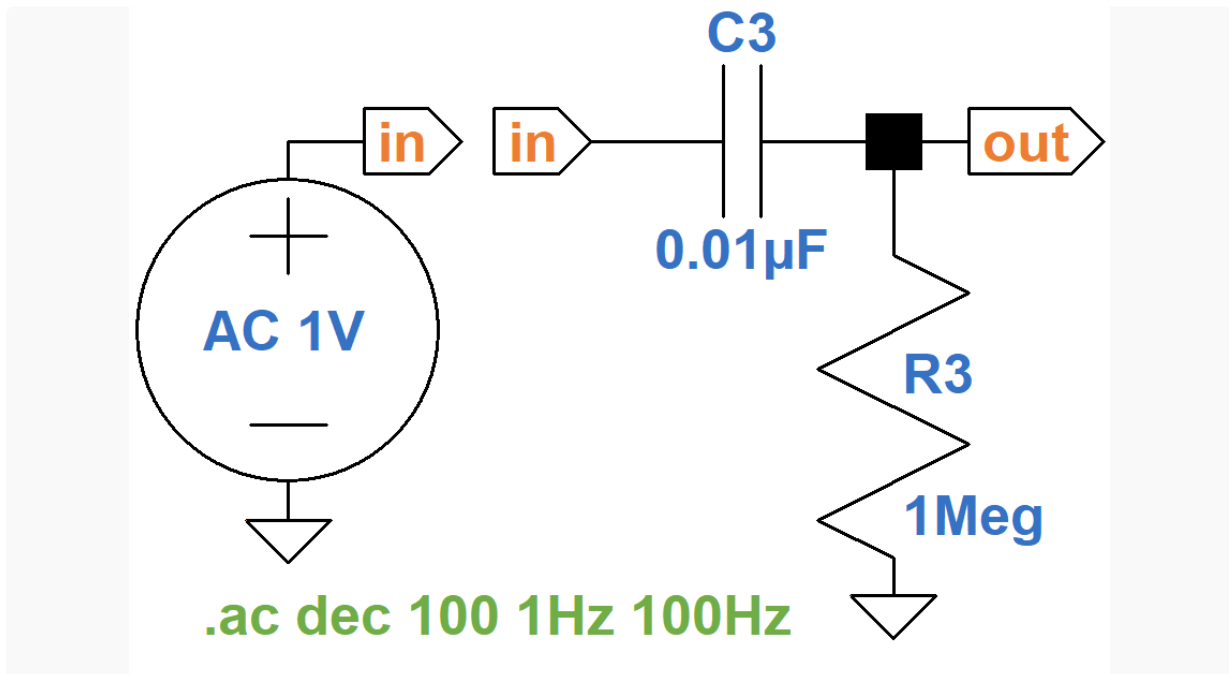
Here is the heart of the Vibroverb 6G16 tremolo circuit: a low-frequency oscillator (LFO) with a footswitch interface and a speed control.



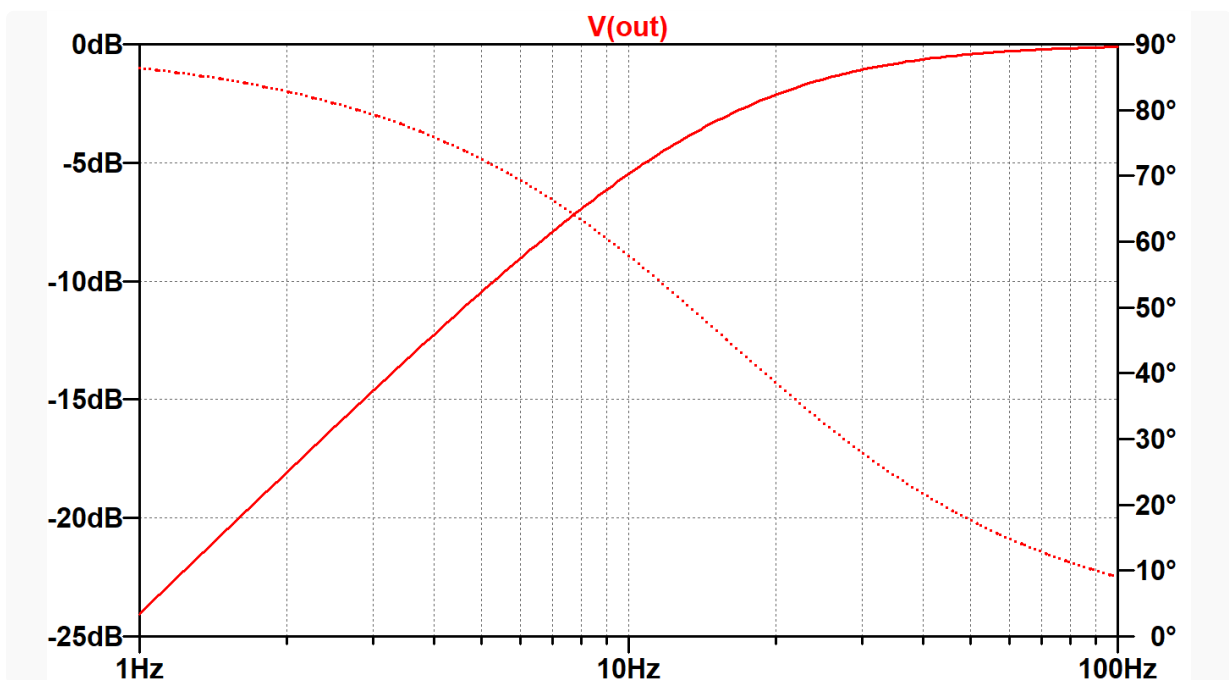
The signal it generates, which has a frequency of a few hertz, is used to modulate the amp's volume, creating a throbbing effect synced to the LFO frequency.

"The tremolo circuit in the 1963 Vibroverb is much simpler than the 'harmonic vibrato' in other large brown Fender amps of the early 1960s, but if you are looking for genuine tremolo (and despite the raves for that brownface vibrato, many players are), this is an extremely good one." -Dave Hunter¹

The circuit to the left of CPK is a classic triode voltage amplifier. It drives a network of resistors and capacitors whose output (node "in") is fed back to the grid for re-amplification. Using LTspice we can set up an AC analysis simulation² of a circuit formed by C3 and R3 in isolation.



The results are typical for an RC high-pass filter.³

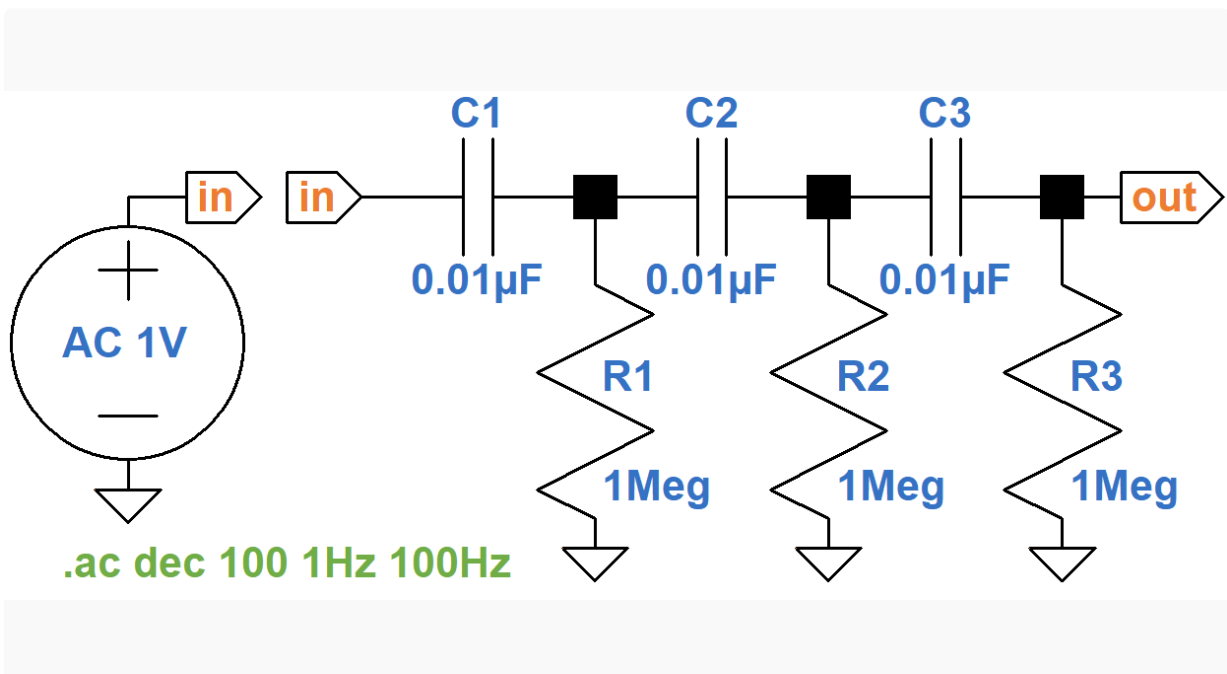


The solid trace for the dB scale at the left shows that at 1Hz there is about 24dB attenuation. At 100Hz there is hardly any attenuation. The dotted trace for the phase angle scale at the right indicates that the phase shift between the input and the output decreases from 86 degrees at 1Hz to only 9 degrees at 100Hz. At about 9Hz the "gain" is -6dB and the phase shift is 60 degrees. This is interesting, because it suggests that if we use three of these high-pass filters in series, we can create a total phase shift of

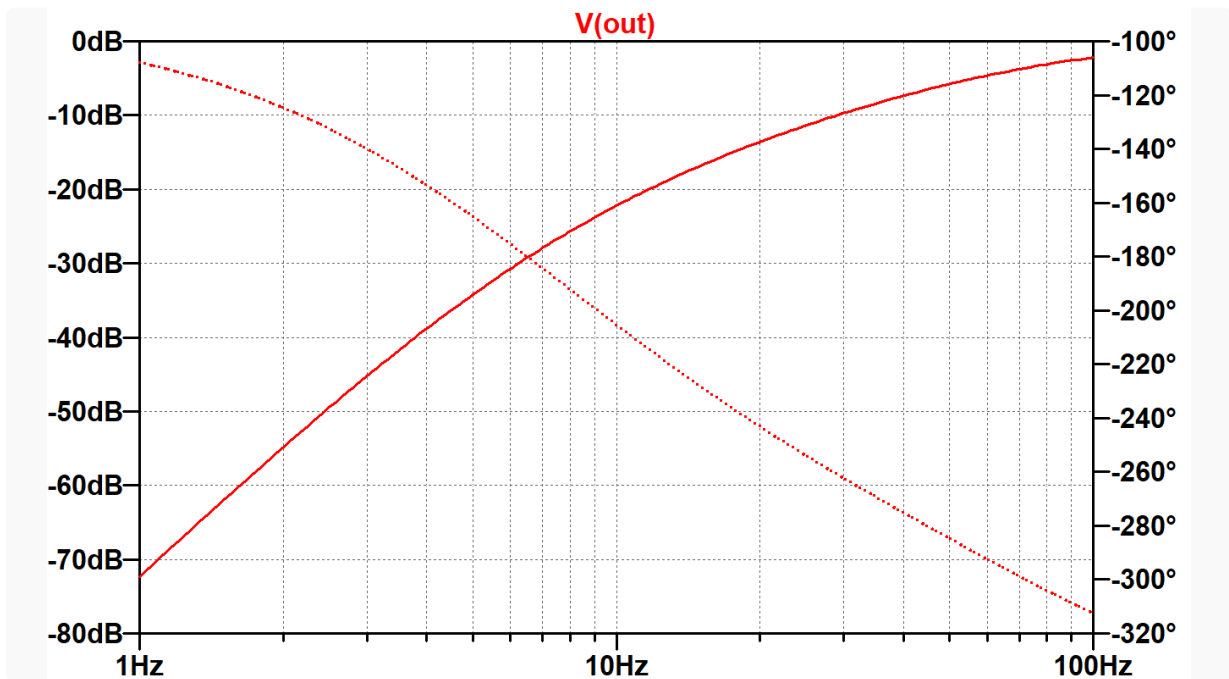
$$60 + 60 + 60 = 180 \text{ degrees}$$

at the cost of about

$6\text{dB} + 6\text{dB} + 6\text{dB} = 18\text{dB}$ attenuation. Let's update the SPICE simulation by adding two more high-pass filters.

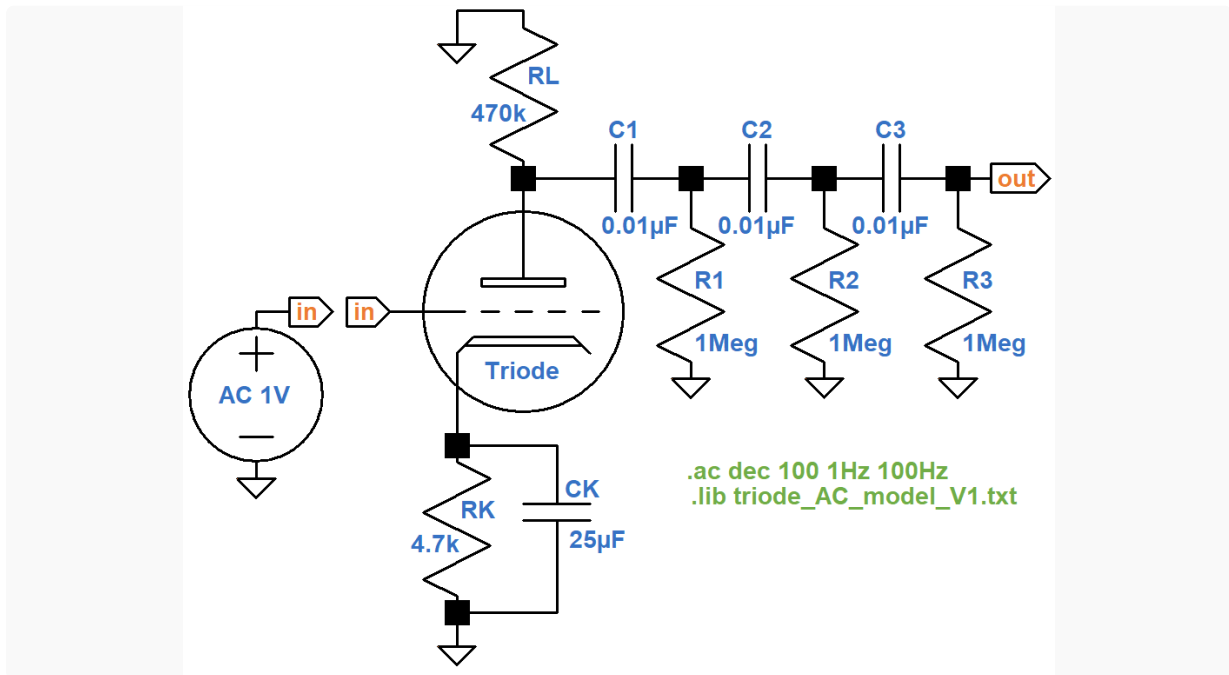


Because the three filters are not independent, the actual frequency for a 180-degree phase shift is near 7Hz, where the "gain" is about -30dB.

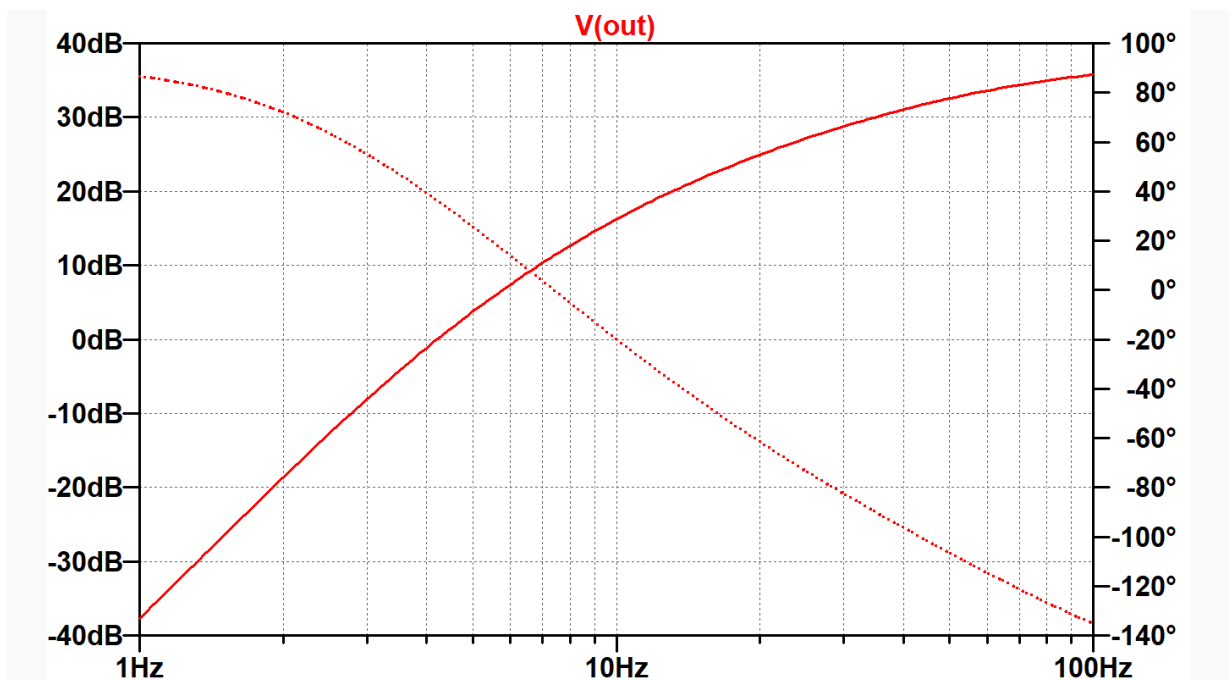


A typical 12AX7 voltage amplifier with a fully bypassed cathode resistor and a light AC load easily creates more than 30dB gain, which can more than compensate for the attenuation of the RC filters. It is also an inverting amplifier - its output is 180 degrees out of phase with its input. Conceivably, if the 3 RC filters are driven by a 12AX7 amplifier, the output will be in phase with the amplifier's input and will have a greater amplitude than the input. Under these conditions, the output can be connected to the 12AX7 grid, creating *positive feedback* that can force the circuit into oscillation.

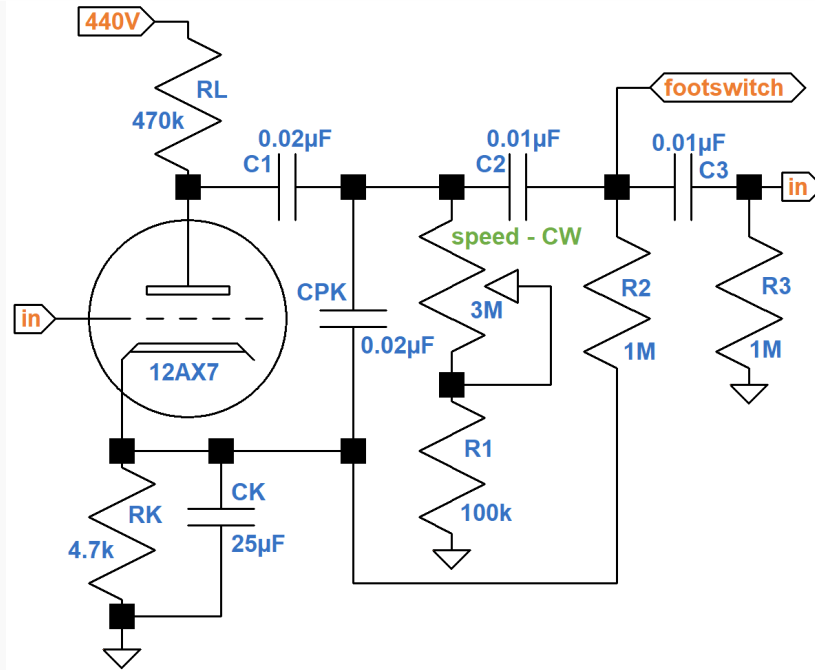
Let's use a 12AX7 AC model4 to drive the circuit.



AC analysis indicates that the output is in phase with the input, i.e. 0 degrees phase shift, at 7.4Hz. The gain at that frequency is +12.3dB.



The Vibroverb circuit adds capacitor CPK between the plate and the cathode to smooth oscillator performance.

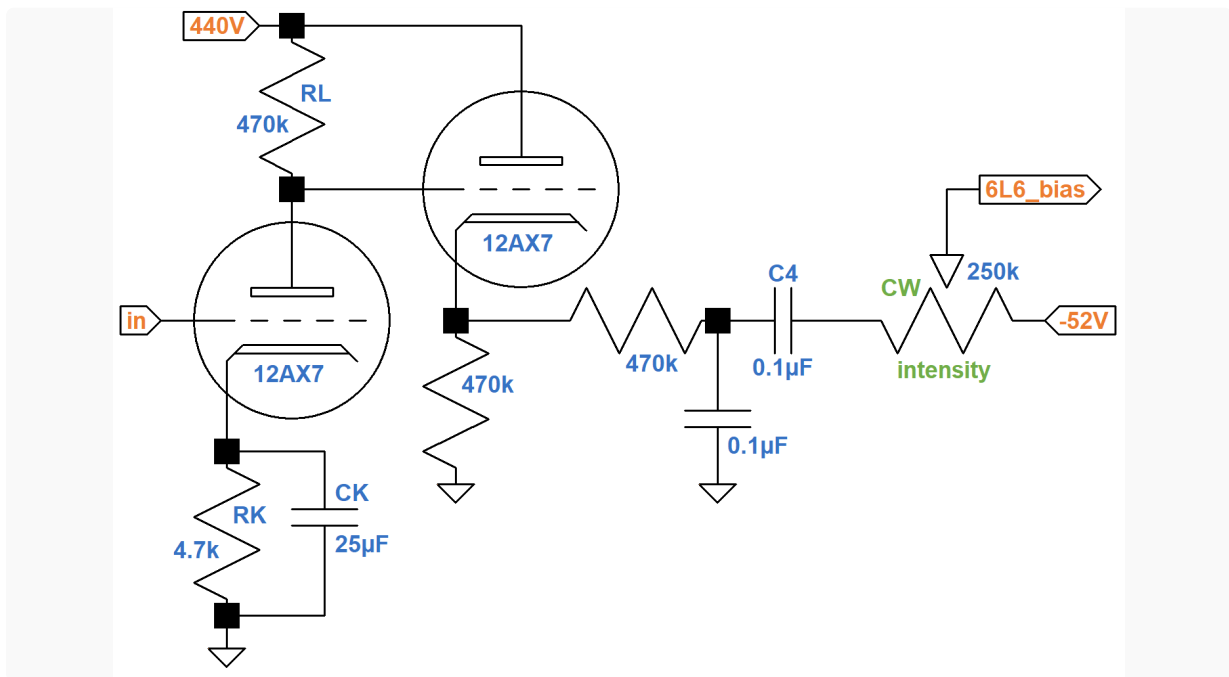


To stop tremolo oscillation, the footswitch shorts the signal at the top of R2 to ground. The bottom of R2 is connected to the cathode instead of ground. Since the cathode is fully bypassed by CK, even at 7Hz, R2 is still connected to AC ground, so oscillation performance is about the same. The cathode voltage is at about 2VDC, however, so when the footswitch opens there is an initial AC jolt to get LFO oscillation started.

Fender also adds a speed control to the first high-pass filter. As the effective value of R1 (the 100kΩ fixed resistor plus the resistance of the potentiometer) decreases, the oscillator frequency increases, thereby driving the tremolo faster.

The Buffer and Bias Modulator

In addition to the three high-pass filters, the triode voltage amplifier also drives a cathode follower that acts as a buffer.



The buffer enables the circuit to drive a heavier AC current load and keeps the load from affecting the LFO frequency. Like the driver for the Fender Bassman tone stack, the cathode follower is DC coupled.⁵

Capacitor C4 blocks DC, so the intensity control sits between a pure AC signal from the LFO and a pure DC voltage from the DC grid bias supply. The bias voltage for the 6L6 power tubes is therefore a combination of the two. When the control is fully clockwise, the -52VDC bias is AC modulated to its maximum extent. Even when cranked, however, negative swings of the bias voltage are short of forcing the amp into Class B operation or beyond. This creates a softer, more nuanced character compared to an opto-cell circuit like the one in the Vibroverb AA763.

"After many years of the seeming ubiquity of the blackface-style opto-cell tremolo circuit, this bias-modulating trem has made a big comeback in popularity recently. Among the characteristics players seem to love are its smooth, watery response - with a soft edge to its rise and fall even at higher intensities - and the way it politely avoids chopping out notes as you play, the way an opto trem can do. Hit the guitar hard and the trem recedes a little, bubbling back into focus as the attack ebbs and the note begins to decay. Many find it the most musical breed of tremolo available, arguably the best type ever built into a tube amp ..." -Dave Hunter⁶

References

1Dave Hunter, **Amped**, (Minneapolis: Voyageur Press, 2012), p. 102.

2Richard Kuehnel, [Guitar Amplifier Electronics: Circuit Simulation](#), (Seattle: Amp Books, 2019), pp. 12-14.

3Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018), pp. 21.

4Richard Kuehnel, [Guitar Amplifier Electronics: Circuit Simulation](#), (Seattle: Amp Books, 2019), pp. 49-64.

5Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018), pp. 84-87.

6Dave Hunter, **Amped**, (Minneapolis: Voyageur Press, 2012), p. 104.